RAZEEN HUSSAIN

VIRTUAL REALITY / AUGMENTED REALITY EXPERT

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CAREER SUMMARY

An experienced researcher specializing in immersive technologies, advanced tracking systems, and human–computer interaction, with a strong record of collaboration across clinical, academic, and industrial domains. Skilled in developing VR/AR applications, designing experimental protocols, and translating complex research challenges into deployable solutions. Motivated to contribute to a forward-thinking organization. Open to relocation.

EDUCATION

PhD in Computer Science 2018 – 2022
University of Genoa, Italy
Thesis Title: Towards Achieving Natural Visual Experience in Immersive Virtual Environments: Addressing Cybersickness and Depth Perception

European Masters on Advanced Robotics 2015 – 2017
University of Genoa, Italy & Warsaw University of Technology, Poland
Thesis Title: Investigation of design methods for developing the control system for walking machine

BE in Mechatronics Engineering 2011 – 2015
National University of Sciences and Technology, Pakistan
Thesis Title: Development of an EEG controlled 6 degree of freedom upper limb prosthesis

WORK EXPERIENCE

Researcher | University of Genoa, Italy 2022 – Present

- Developed VR applications for body-image assessment, integrating psychometric requirements with real-time interaction design.
- Collaborated with French research partners to evaluate the clinical viability of photorealistic avatars (e.g., MetaHuman) for assessment and therapeutic use.
- Investigated the impact of graphics realism on social cognition, designing controlled experiments to isolate perceptual and behavioral effects.
- Co-designed AR-assisted surgical support tools with industrial partners to aid surgeons during cochlear implant procedures.
- Led the development of empathic virtual agents for student-counseling scenarios, integrating affective computing and conversational interaction.
- Designed and built serious games for Alzheimer's assessment in collaboration with clinical psychologists, deploying cross-platform builds (VR, Android, WebGL) using Unity.
- Developed a custom OCR pipeline for Microsoft HoloLens 2, leveraging device-specific spatial mapping and AR interaction capabilities.
- Researched VR locomotion techniques with a focus on improving distance-perception accuracy and reducing perceptual bias.

- Applied photogrammetry workflows to reconstruct high-fidelity 3D assets for seamless integration into AR/VR environments.
- Collaborated with German research partners to evaluate spatial blurring techniques in passive and active VR contexts.
- Assessed multiple foveation strategies to determine their impact on perceptual naturalness and user comfort.
- Provided comprehensive supervision to multiple undergraduate and master's students, specifically focusing on research and development of serious games and VR applications.

Consultant | Freelance 2017– 2024

- Collaborated as part of a technology scouting team to identify and evaluate the latest advancements in software and hardware technologies relevant to XR, AI and robotics.
- Contributed to the development of a VR-based remote operator training application, incorporating software and hardware components to create a robust immersive training environment.
- Designed and implemented a vision-based algorithm for aerial detection of high-tension wires, utilizing software development skills to enhance safety measures.
- Led the development of a software architecture for a domestic cleaning robot, ensuring efficient integration and interaction between various software and hardware components.

PhD Researcher | University of Genoa, Italy 2018 – 2022

- Led a research project focused on advancing techniques to modify visual stimuli in immersive media devices, specifically targeting the improvement of cybersickness and depth perception. This involved exploring innovative approaches and conducting experiments to validate the effectiveness of the proposed solutions.

Supply Chain Intern | Unilever Pakistan Limited, Pakistan 2014

- Designed a comprehensive machine ledger for the Viking Cup Dispenser, creating an organized and accessible system for tracking and documenting machine maintenance activities.
- Conducted critical spare parts analysis for the production line, identifying key components that significantly impact the reliability and uptime of the machinery.

Engineering Intern | Pakistan Oilfields Limited, Pakistan 2013

- Contributed to the maintenance of natural gas generators, Variable Frequency Drives (VFDs), and Programmable Logic Controllers (PLCs) at Pariwali Production Facilities.
- Actively participated in scheduled maintenance activities, including inspections, cleaning, and testing of equipment to prevent potential failures and maintain optimal performance.

PROFESSIONAL DEVELOPMENT

- Italian Language A2 Certification | CPIA Levante Tigullio | Genoa, Italy 2026
- Data Science Summer School (DS3) | Hertie School Data Science | Online 2023
- International Computer Vision Summer School (ICVSS) | University of Catania | Sicily, Italy 2019
- Machine Learning Crash Course (MLCC) | University of Genoa | Genoa, Italy 2019
- Regularization Methods for Machine Learning (RegML) | University of Genoa | Genoa, Italy 2018
- Machine Learning | Stanford University on Coursera | Online 2018

LANGUAGES

English (C2) | Italian (A2) | Polish (A1) | Urdu (Native)

TECHNICAL SKILLS

- **Languages & Engines:** C/C++, C#, Python, MATLAB, Unity, Unreal Engine, WebGL, Cg/HLSL
- **XR Hardware & SDKs:** Microsoft Hololens 2, HTC Vive, Meta Quest, SteamVR, MRTK, XR Interaction Toolkit
- **Computer Vision & ML:** OpenCV, TensorFlow, PyTorch, Photogrammetry, Gaze Tracking, Hand Tracking, Haptics
- **Tools & Other:** Git, Blender, MeshLab, ROS, Creo, SolidWorks, EyesWeb, Microsoft Office

AWARDS

- 1st position in EndoVis-SurgVisDom Challenge (Category II) at MICCAI 2020
- President's Gold Medal for Best in Academics at NUST CEME Graduation Ceremony 2015
- Rector's Gold Medal for Best Final Year Project at NUST CEME Graduation Ceremony 2015

PUBLICATIONS

Journal Articles

- **Razeen Hussain**, Manuela Chessa & Fabio Solari (2026). Gaze-contingent Focus Cues to Enhance Depth Perception in Mixed Reality Environments. Virtual Reality, 30(4). DOI: 10.1007/s10055-026-01438-8
- **Razeen Hussain**, Manuela Chessa & Fabio Solari (2026). Comparing Distance Perception Across Common Virtual Reality Locomotion Methods. Multimedia Tools and Applications, 85(7). DOI:10.1007/s11042-026-21747-z
- **Razeen Hussain**, Muhammad Mohsin, Dadan Khan & Mohammad Zohaib (2026). A Systematic Review of Deep Learning Approaches for Hepatopancreatic Tumor Segmentation. Journal of Imaging, 12(4). DOI:10.3390/jimaging12040147
- Shayan Alvansazyazdi, **Razeen Hussain**, Jan Margeta, Manuela Chessa, Roger Calixto, Reda Abdellah Kamraoui, Raabid Hussain & Fabio Solari (2026). Vision-Based Object Detection and Tracking from Surgical Microscope Video for Augmented Cochlear Implant Surgery. Cochlear Implants International, 27(2). DOI:10.1080/14670100.2026.2622144
- Valeria Manera, Maria Cordero-Rull, **Razeen Hussain** & Fabio Solari (2025). The interest of eye-tracking for apathy detection in patients with Mild Neurocognitive Disorders. Alzheimer's & Dementia, 21:e099728. DOI:10.1002/alz70856_099728
- **Razeen Hussain**, Manuela Chessa & Fabio Solari (2023). Improving depth perception in immersive media devices by addressing vergence-accommodation conflict. IEEE Transactions on Visualization & Computer Graphics (TVCG), 30(9). DOI:10.1109/TVCG.2023.3331902
- **Razeen Hussain**, Manuela Chessa & Fabio Solari (2021). Mitigating cybersickness in virtual reality systems through foveated depth-of-field blur. Sensors, 21(12). DOI:10.3390/s21124006
- Aneeq Zia, Kiran Bhattacharyya, Xi Liu, Ziheng Wang, Satoshi Kondo, Emanuele Colleoni, Beatrice van Amsterdam, **Razeen Hussain**, Raabid Hussain, Lena Maier-Hein, Danail Stoyanov, Stefanie Speidel & Anthony Jarc (2021). Surgical visual domain adaptation: Results from the MICCAI 2020 SurgVisDom challenge. ArXiv:2102.13644
- **Razeen Hussain**, Teresa Zielinska & Rene Hexel (2019). Finite state automaton based control system for walking machines. International Journal of Advanced Robotic Systems (IJARS), 16(3). DOI:10.1177/1729881419853182

Conference Proceedings

- Celeste Bazzi, **Razeen Hussain**, Manuela Chessa & Fabio Solari (2026). Comparing 3D Gaussian Splatting and Cartoon Style for Intelligent Avatars in Virtual Reality Interactions. In 2026 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct) [IN PRESS]

- Marianna Pizzo, Eros Viola, **Razeen Hussain**, Fabio Solari & Manuela Chessa (2026). From Substitutional Reality to Rehabilitation: Passive-Haptic Chair Registration as a Foundation for Virtual Reality Motor Assessment and Training. In 2026 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct) [IN PRESS]
- **Razeen Hussain**, Manuela Chessa & Fabio Solari (2026). An Overview of Emerging Trends in XR-Enabled Human–AI Collaboration for Circular Economy and Recycling Systems. In 2026 IEEE 10th Forum on Research and Technologies for Society and Industry Innovation (RTSI). [IN PRESS]
- Federica Bruno, **Razeen Hussain**, Manuela Chessa, Guillaume Sacco, Valeria Manera, Maël Addoum & Fabio Solari (2026). Are Digital Characters Suitable for Emotion Recognition Tasks? An Evaluation Study Using MetaHumans. In 2026 ACM 1st International Conference on Human-Computer Interaction in the Alps (AlpCHI). DOI:10.1145/3780045.3780054
- Seyedbehnam Rad, **Razeen Hussain**, Manuela Chessa & Fabio Solari (2026). Designing Emotionally Intelligent Embodied Agents for Immersive Virtual Reality Experiences. In 2026 IEEE 8th International Conference on Artificial Intelligence & extended and Virtual Reality (AIxVR). DOI:10.1109/AIxVR67263.2026.00023
- **Razeen Hussain**, Manuela Chessa & Fabio Solari (2025). A Comparative Study on Locomotion Methods and Distance Perception in Immersive Virtual Reality. In 2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW). DOI:10.1109/VRW66409.2025.00275
- Cristina Díez Bort, **Razeen Hussain**, Valeria Manera, Manuela Chessa & Fabio Solari (2025). A Serious Game for Early Detection and Assessment of Social Apathy: A Pilot Study. In 2025 9th International Conference on Human Computer Interaction Theory and Applications (HUCAPP). DOI:10.5220/0013119700003912
- Manuela Chessa, Lorenzo Gerini, **Razeen Hussain**, Matteo Martini, Marianna Pizzo, Fabio Solari & Eros Viola (2024). Extended Reality and Artificial Intelligence for Exergaming: Opportunities and Open Challenges for Rehabilitation and Cognitive Training. In 2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI). DOI:10.1109/RTSI61910.2024.10761257
- **Razeen Hussain**, Manuela Chessa & Fabio Solari (2024). Exploring foveation techniques for virtual reality environments. In 2024 19th International Conference on Computer Graphics Theory and Applications (GRAPP). DOI:10.5220/0012458200003660
- **Razeen Hussain**, Marianna Pizzo, Giorgio Ballestin, Manuela Chessa & Fabio Solari (2022). Experimental validation of photogrammetry based 3D reconstruction software. In 2022 IEEE 5th International Conference on Image Processing Applications and Systems (IPAS). DOI:10.1109/IPAS55744.2022.10053055
- **Razeen Hussain**, Manuela Chessa & Fabio Solari (2020). Modelling foveated depth-of-field blur for improving depth perception in virtual reality. In 2020 IEEE 4th International Conference on Image Processing, Applications and Systems (IPAS). DOI:10.1109/IPAS50080.2020.9334947
- **Razeen Hussain**, Fabio Solari & Manuela Chessa (2019). Simulated foveated depth-of-field blur for virtual reality systems. In 2019 16th ACM SIGGRAPH European Conference on Visual Media Production (CVMP). DOI:10.13140/RG.2.2.35437.31208
- **Razeen Hussain**, Mustafa Arsal Shahid, Jibran Ahmed Khan, Mohsin Islam Tiwana, Javaid Iqbal & Nasir Rashid (2015). Development of a low-cost anthropomorphic manipulator for commercial usage. In 2015 IEEE International Symposium on Innovations in Intelligent SysTems and Applications (INISTA). DOI:10.1109/INISTA.2015.7276750

Book Chapters

- Raabid Hussain, Shayan Alvansazyazdi, Giuseppe Ricciardi & **Razeen Hussain** (2026). Advanced Surgical Techniques Using Robotics and Augmented Reality. In Behnam Molaee & Manuel Segovia (Eds.), Advancing Cochlear Implants: Current and Emerging Methods and Technologies. DOI:10.1007/978-3-032-15719-5_14

Patents

- **Razeen Hussain**, Fabio Solari & Manuela Chessa (2023). Method based on inverse blurring for immersive media devices to mitigate vergence accommodation conflict. PCT/IB2023/059732, filed 29/09/2023. [WIPO Patent Pending]
- **Razeen Hussain**, Fabio Solari & Manuela Chessa (2022). Metodo basato sulla sfocatura inversa per dispositivi multimediali immersivi per attenuare il conflitto vergenza/accomodamento. IT102022000022278, filed 28/10/2022, granted 10/10/2024.

REVIEWING ACTIVITIES

Journals

- IEEE Transactions on Visualization and Computer Graphics (TVCG)
- Springer Virtual Reality
- IEEE Transactions on Human-Machine Systems (THMS)
- Frontiers in Virtual Reality
- Springer Multimedia Systems
- IEEE Computer Graphics and Applications (CG&A)
- International Journal of Human-Computer Interaction (IJHCI)
- International Journal of Industrial Ergonomics (IJIE)

Conferences

- IEEE International Symposium on Mixed and Augmented Reality (ISMAR) | 2026 | 2025 | 2024
- IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) | 2026 | 2025
- ACM CHI Conference on Human Factors in Computing Systems (CHI) | 2026 | 2025
- ACM Conference on Intelligent User Interfaces (IUI) | 2027 | 2026
- IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) | 2026 | 2024
- International Conference of the Immersive Learning Research Network (iLRN) | 2025 | 2024
- ACM International Conference on Interactive Media Experiences (IMX) | 2025
- International Conference on Artificial Intelligence and Virtual Reality (AIVR) | 2024
- International Workshop on Assistive Computer Vision and Robotics (ACVR) | 2023
- IEEE International Conference on Image Processing Applications and Systems (IPAS) |2022

TEACHING ACTIVITIES

Computer Vision | MSc Robotics Engineering | University of Genoa, Italy

- Academic Years: 2025/2026 | 2024/2025 | 2023/2024 | 2022/2023 | 2021/2022 | 2020/2021 | 2019/2020

Algorithms & Object Oriented Programming | BSc Biomedical Engineering | University of Genoa, Italy

- Academic Years: 2025/2026 | 2024/2025 | 2023/2024 | 2022/2023 | 2021/2022 | 2020/2021

Augmented Reality | MSc Computer Science | University of Genoa, Italy

- Academic Years: 2023/2024

SUPERVISING ACTIVITIES

University of Genoa, Italy 2021 – Present

- MSc Computer Science | 5 Students
- MSc Bioengineering | 3 Students
- MSc Design Engineering | 1 Student
- BSc Biomedical Engineering | 11 Students
- BSc Computer Science | 1 Student

SEMINARS/TALKS

- An Overview of Emerging Trends in XR-Enabled Human-AI Collaboration for Circular Economy and Recycling Systems, August 18, 2026, Helsinki, Finland
- Shaders in Unity, May 12, 2026, Genoa, Italy
- Are Digital Characters Suitable for Emotion Recognition Tasks? An Evaluation Study Using MetaHumans, March 3, 2026, Ascona, Switzerland
- Designing Emotionally Intelligent Embodied Agents for Immersive Virtual Reality Experiences, January 28, 2026, Osaka, Japan
- Rendering & Shaders in Unity, May 19, 2025, Genoa, Italy
- A Serious Game for Early Detection and Assessment of Social Apathy: A Pilot Study, February 27, 2025, Porto, Portugal
- Perception & Interaction Lab @ DIBRIS, December 11, 2024, Genoa, Italy
- Exploring Foveation Techniques for Virtual Reality Environments, February 28, 2024, Rome, Italy
- Perceptual Issues in Virtual Reality Systems, April 22 , 2021, Genoa, Italy
- Modelling Foveated Depth-of-field Blur for Improving Depth Perception in Virtual Reality, December 10, 2020, Genoa, Italy
- Defocus Blur in Virtual Environments, April 23, 2020, Genoa, Italy
- Simulated Foveated Depth-of-field Blur for Virtual Reality Systems, December 18, 2019, London, UK
- Visual Perception in Virtual Reality Systems, October 17, 2019, Genoa, Italy

REFERENCES

- Prof. Fabio Solari
  Associate Professor | University of Genoa | fabio.solari@unige.it

- Prof. Manuela Chessa
  Associate Professor | University of Genoa | manuela.chessa@unige.it